Execution Methods

2. What do we do with Spark?

---

1. How to create Spark Programs?
2. How to execute Spark Programs?

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How to execute Spark Programs?

udemy

Execution Methods

1. Interactive Clients
2. Submit Job

Execution Methods

APACHE Spark™

1. Interactive Clients

spark-shell, Notebook

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Execution Methods

Stream Processing

APACHE Spark

Batch Processing

YouTube

APACHE Spark

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Execution Methods

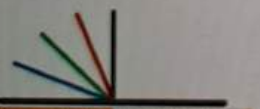
2. Submit Job

spark-submit

Execution Methods

How to execute Spark Programs?

1. Interactive Clients
   spark-shell, Notebook
2. Submit Job
   spark-submit, Databricks Notebook, Rest API



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Processing Model

Spark Client

Spark-submit APP

Spark Cluster

Cluster Manager (YARN)

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Spark Distributed Processing Model?

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Processing Model

Spark Client

Spark-submit

Spark Cluster

Cluster Manager (YARN)

Driver

Executor

Executor

Executor

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Processing Model

Spark Client

Spark-submit A2

Spark Cluster

Cluster Manager (YARN)

A1 Driver

A1 Executor

A1 Executor

A1 Executor

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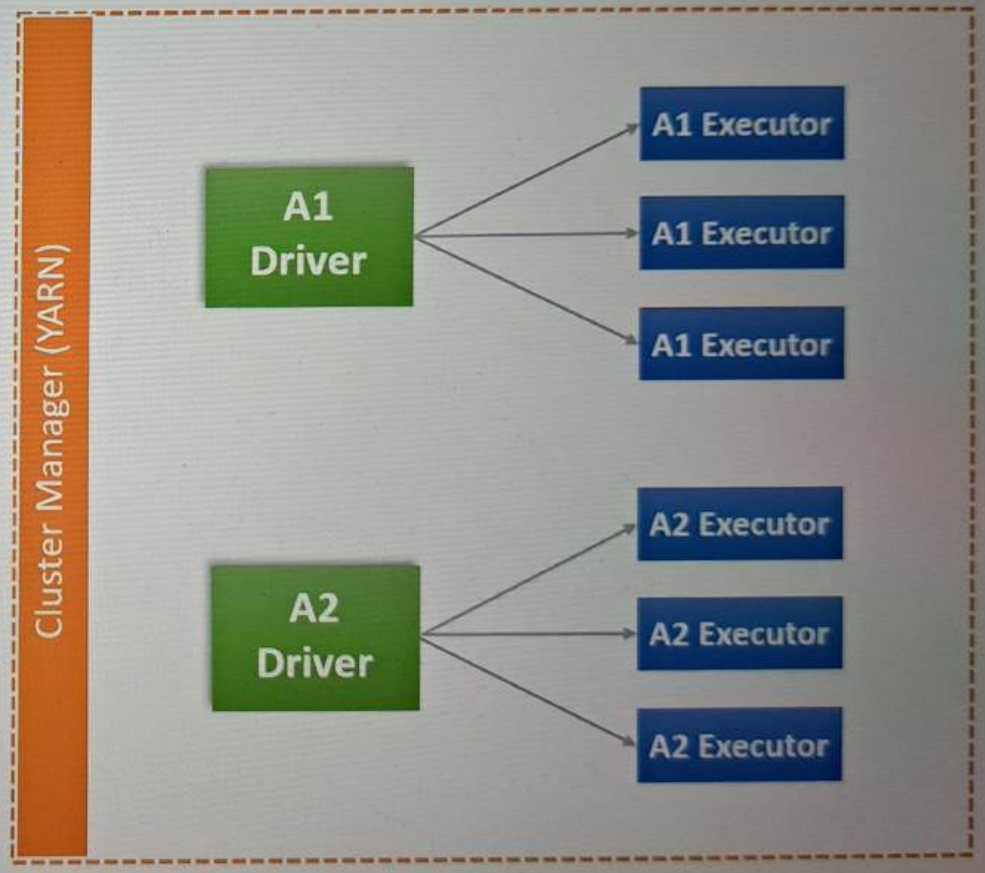
Processing Model

Spark Client



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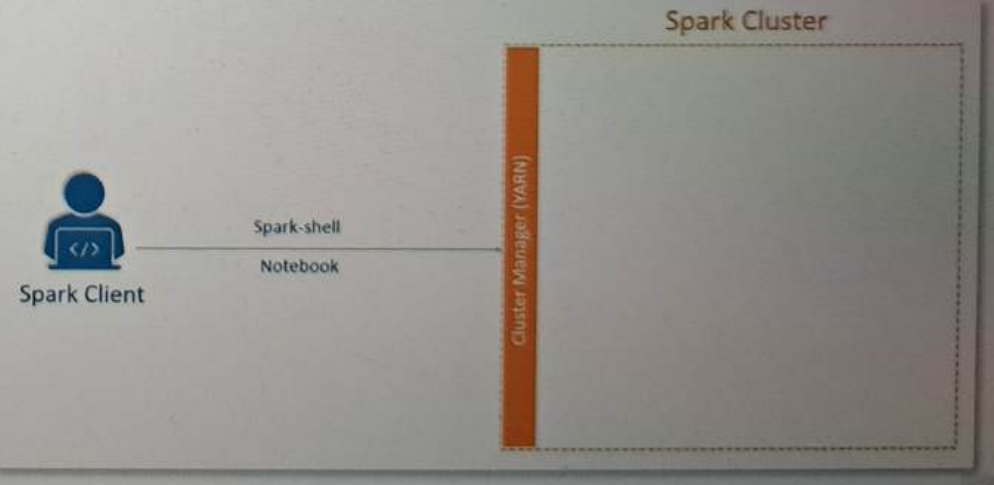
Spark local & Execution Modes

Spark-shell | Notebook | IDE

How Spark works on Local Machine?

How Spark works with Interactive Clients?



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How to Spark on local machine?

Udemy

File Edit View Navigate Code Refactor Run Tools VCS Window Help 01-HelloSpark - HelloSpark.py - PyCharm

01-HelloSpark > HelloSpark.py

Unittests for test_utile.UtilsTestCase

Project

- 01-HelloSpark C:\Users\prashant\Documents\GitHub\Spar
  - app-logs
  - data
  - lib
  - HelloSpark.py
  - log4j.properties
  - requirements.txt
  - spark.conf
  - test_utile.py
- External Libraries
- Scratches and Consoles

HelloSpark.py

```python
import sys
from pyspark.sql import *
from lib.logger import Log4j
from lib.utils import *

if __name__ == "__main__":
    conf = get_spark_app_config()

    spark = SparkSession \
        .builder \
        .appName("HelloSpark") \
        .master("local[2]") \
        .getOrCreate()

    logger = Log4j(spark)

    if len(sys.argv) != 2:
        logger.error("Usage: HelloSpark <filename>")
        sys.exit(-1)

    logger.info("Starting HelloSpark")

    survey_raw_df = load_survey_df(spark, sys.argv[1])
    partitioned_survey_df = survey_raw_df.repartition(2)
    count_df = count_by_country(partitioned_survey_df)
    count_df.show()

    logger.info("Finished HelloSpark")
    spark.stop()
```

Git TODO Terminal Python Console

Event Log

5:1 CRLF UTF-8 4 spaces Python 3.7 (01-HelloSpark) master

Spark Cluster Managers

1. local[n]
2. YARN
3. Kubernetes
4. Mesos
5. Standalone

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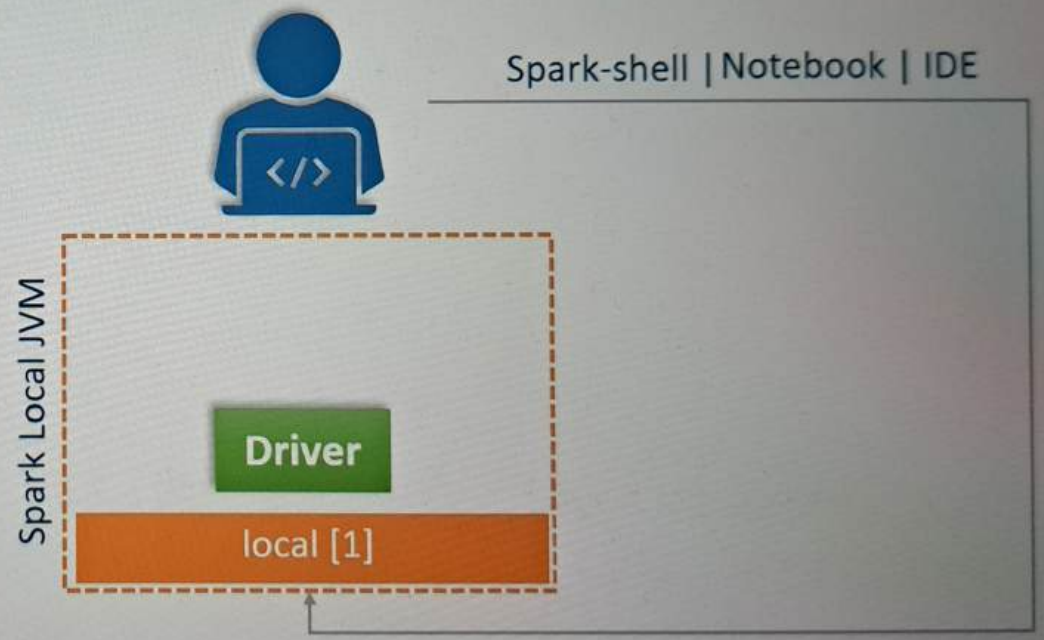
Spark Local

1. local[n]
2. YARN
3. Kubernetes
4. Mesos
5. Standalone



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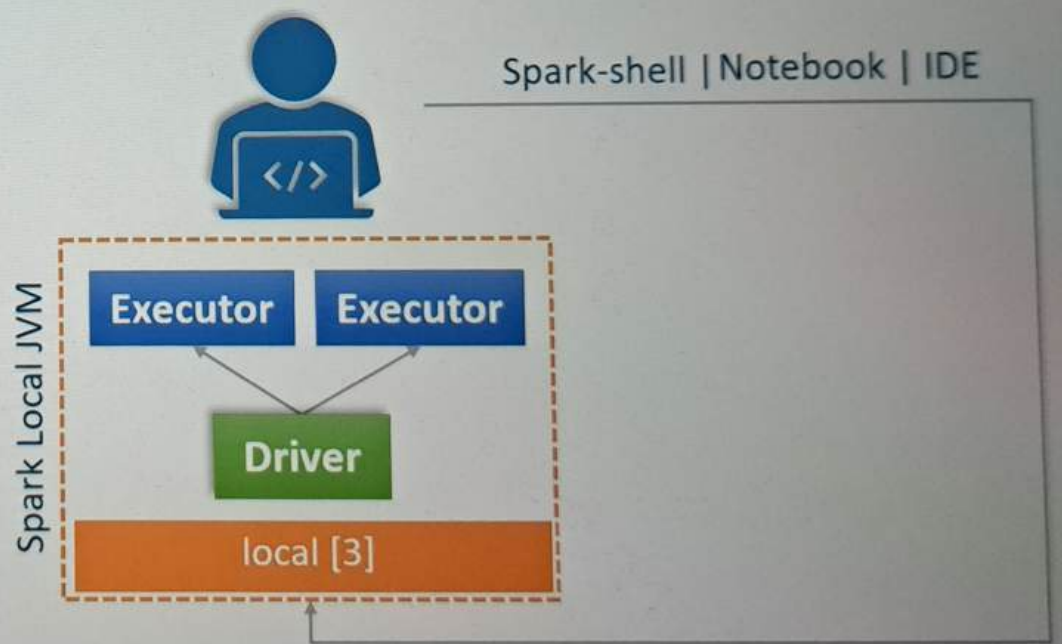
Spark Local

1. local[n]
2. YARN
3. Kubernetes
4. Mesos
5. Standalone



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How to Spark with interactive clients?

udemy

Spark Execution Modes

1. Client
2. Cluster

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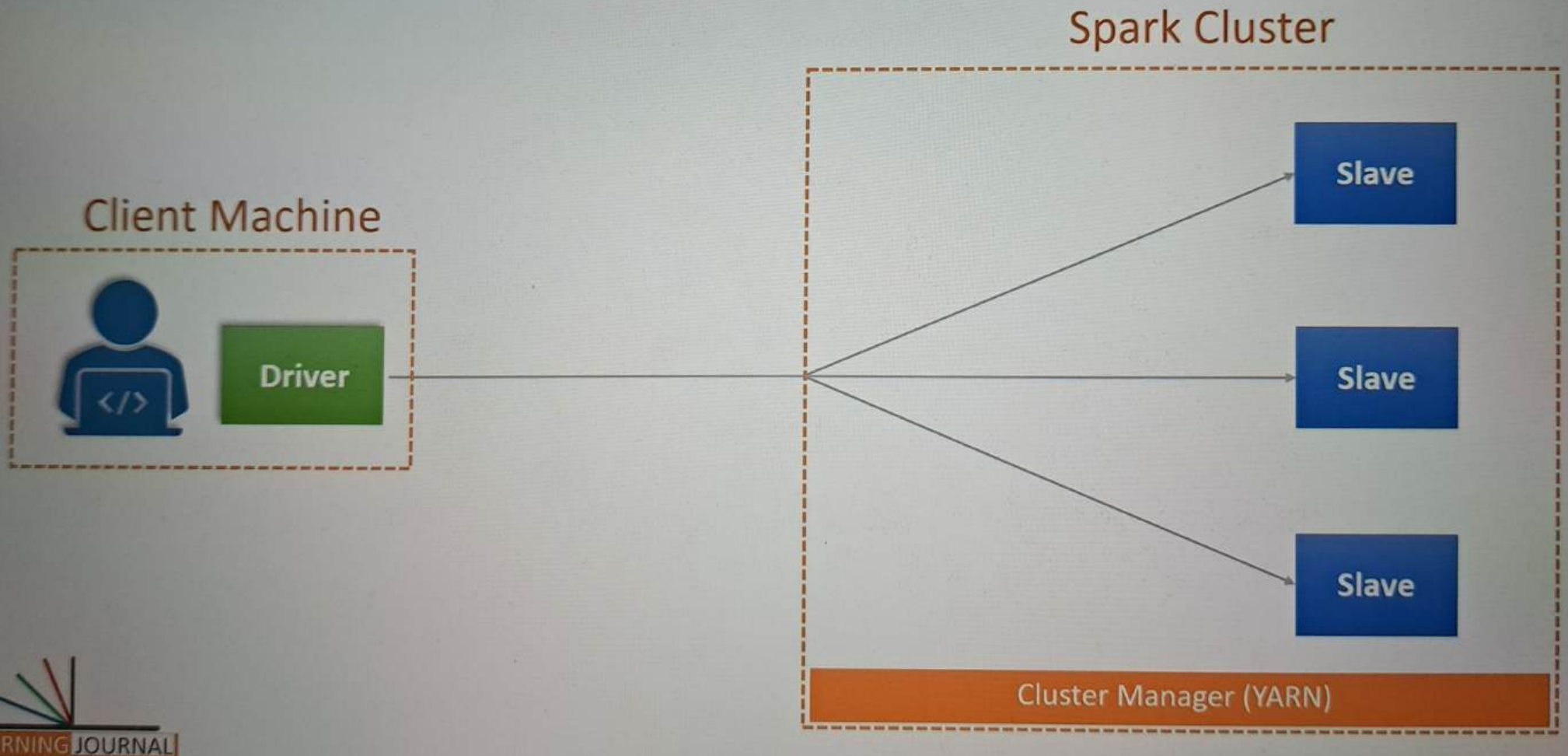
Client Mode



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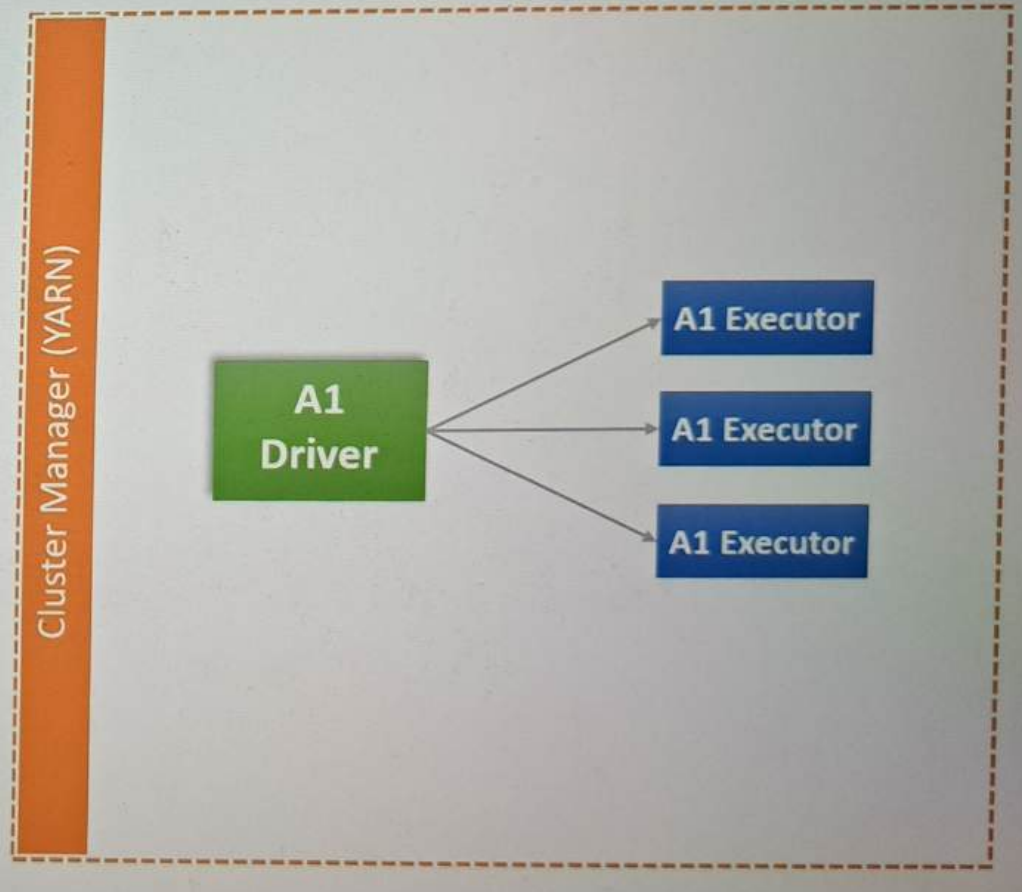
Processing Model

Spark Client



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Spark Execution Model

Cluster Managers	Execution Modes	Execution Tools
1. local[n]	1. Client	1. IDE, Notebook
2. YARN		
3. Kubernetes	2. Cluster	2. Submit
4. Mesos		
5. Standalone		

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Spark Execution Model

Cluster Managers | **Execution Modes** | Execution Tools

1. local[n]
2. YARN
   - On-Premise
   - On-Cloud

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Spark Execution Model

Cluster Managers	Execution Modes	Execution Tools
1. local[n]	1. Client	1. IDE, Notebook
2. YARN	2. Client	2. Notebook, Shell
3. YARN	3. Cluster	3. Spark Submit

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PySparkShell - Spark Jobs

localhost:4040/jobs/

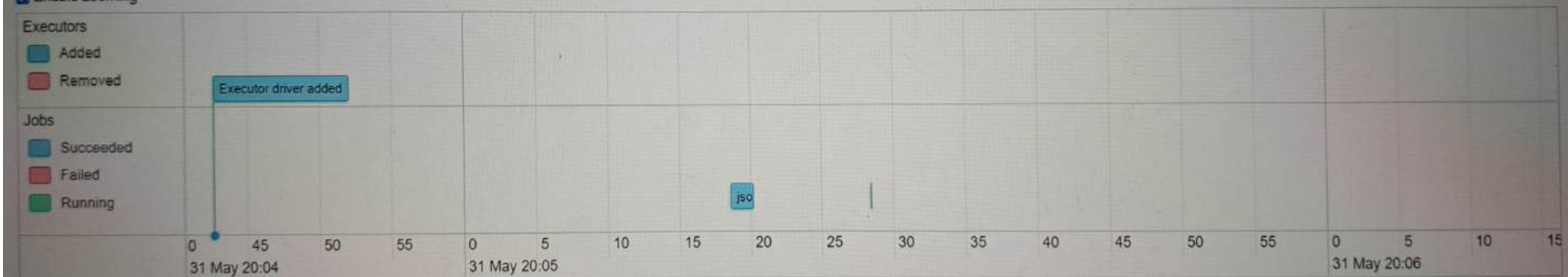
Spark 3.0.0-preview2 | Jobs | Stages | Storage | Environment | Executors | SQL | PySparkShell application UI

Spark Jobs (?)

User: prashant
Total Uptime: 1.6 min
Scheduling Mode: FIFO
Completed Jobs: 2

Event Timeline

Enable zooming



Completed Jobs (2)

Page: 1

1 Pages. Jump to 1 . Show 100 items in a page. Go

Job Id	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
1	showString at NativeMethodAccessorImpl.java:0 showString at NativeMethodAccessorImpl.java:0	2020/05/31 20:05:28	0.1 s	1/1	1/1
0	json at NativeMethodAccessorImpl.java:0 json at NativeMethodAccessorImpl.java:0	2020/05/31 20:05:18	2 s	1/1	1/1

PySparkShell - Executors

localhost:4040/executors/

Spark 3.0.0-preview2 | Jobs | Stages | Storage | Environment | Executors | SQL | PySparkShell application UI

Executors

Show Additional Metrics
- Select All
- On Heap Memory
- Off Heap Memory
- Resources

Summary

	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Blacklisted
Active(1)	0	29.5 KiB / 912.3 MiB	0.0 B	3	0	0	2	2	1 s (0.0 ms)	103 B	0.0 B	0.0 B	0
Dead(0)	0	0.0 B / 0.0 B	0.0 B	0	0	0	0	0	0.0 ms (0.0 ms)	0.0 B	0.0 B	0.0 B	0
Total(1)	0	29.5 KiB / 912.3 MiB	0.0 B	3	0	0	2	2	1 s (0.0 ms)	103 B	0.0 B	0.0 B	0

Executors

Show 20 entries

Search:

Executor ID	Address	Status	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Thread Dump
driver	prashant-W541.mshome.net:51070	Active	0	29.5 KiB / 912.3 MiB	0.0 B	3	0	0	2	2	1 s (0.0 ms)	103 B	0.0 B	0.0 B	Thread Dump

Showing 1 to 1 of 1 entries

Previous 1 Next

prashant@yarn-cluster-m: ~ - Google Chrome

ssh.cloud.google.com/projects/prashant-gcp-project/zones/us-central1-b/instances/yarn-cluster-m?authuser=1&hl=en_GB&projectNumber=315784728289&useAdmin...

```
prashant@yarn-cluster-m:~$
prashant@yarn-cluster-m:~$
prashant@yarn-cluster-m:~$
prashant@yarn-cluster-m:~$
prashant@yarn-cluster-m:~$ pyspark --master yarn --driver-memory 1G --executor-memory 500M --num-executors 2 --executor-cores 1
```

Cluster details – Dataproc – Pras

https://ebnrzrzk3jfkbb6dujpuicu

ebnrzrzk3jfkbb6dujpuicuvgi-dot-us-central1.dataproc.googleusercontent.com/gateway/default/sparkhistory/history/application_1591009784507_0001/jobs/

prashant-gcp-project > yarn-cluster

Sign out

Spark 2.4.5

Jobs | Stages | Storage | Environment | Executors

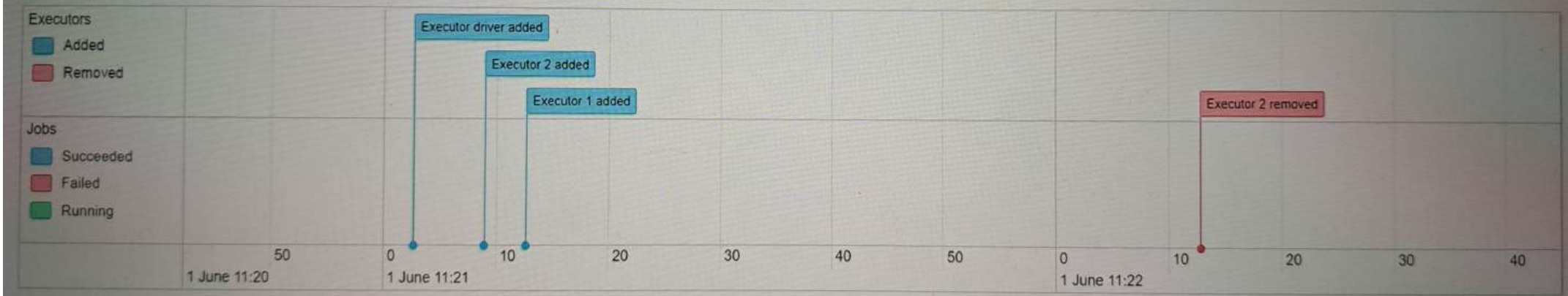
PySparkShell application UI

Spark Jobs (?)

User: prashant
Total Uptime: 8.9 min
Scheduling Mode: FAIR

▾ Event Timeline

☑ Enable zooming



Waiting for ebnrzrzk3jfkbb6dujpuicuvgi-dot-us-central1.dataproc.googleusercontent.com...

Cluster details – Dataproc – Pras | https://ebnrzrzk3jfkbb6dujpuicu

ebnrzrzk3jfkbb6dujpuicuvgi-dot-us-central1.dataproc.googleusercontent.com/gateway/default/sparkhistory/history/application_1591009784507_0001/executors/

prashant-gcp-project > yarn-cluster

Sign out

Spark 2.4.5 | Jobs | Stages | Storage | Environment | Executors | PySparkShell application UI

Executors

Show Additional Metrics

Summary

	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Blacklisted
Active(2)	0	0.0 B / 499.5 MB	0.0 B	1	0	0	0	0	0 ms (0 ms)	0.0 B	0.0 B	0.0 B	0
Dead(1)	0	0.0 B / 115.4 MB	0.0 B	1	0	0	0	0	0 ms (0 ms)	0.0 B	0.0 B	0.0 B	0
Total(3)	0	0.0 B / 614.8 MB	0.0 B	2	0	0	0	0	0 ms (0 ms)	0.0 B	0.0 B	0.0 B	0

Executors

Show 20 entries

Search:

Executor ID	Address	Status	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Logs
driver	yarn-cluster-m.c.prashant-gcp-project.internal:37559	Active	0	0.0 B / 384.1 MB	0.0 B	0	0	0	0	0	0 ms (0 ms)	0.0 B	0.0 B	0.0 B	
1	yarn-cluster-w-0.c.prashant-gcp-project.internal:44187	Active	0	0.0 B / 115.4 MB	0.0 B	1	0	0	0	0	0 ms (0 ms)	0.0 B	0.0 B	0.0 B	stdout stderr
2	yarn-cluster-w-1.c.prashant-gcp-project.internal:38825	Dead	0	0.0 B / 115.4 MB	0.0 B	1	0	0	0	0	0 ms (0 ms)	0.0 B	0.0 B	0.0 B	stdout stderr

Showing 1 to 3 of 3 entries

Previous 1 Next

Cluster details – Dataproc – Pras | https://ebnrzrzk3jfkbb6dujpuicu | https://ebnrzrzk3jfkbb6dujpuicu

ebnrzrzk3jfkbb6dujpuicuvgi-dot-us-central1.dataproc.googleusercontent.com/gateway/default/yarn/proxy/application_1591009784507_0002/environment/

prashant-gcp-project > yarn-cluster Sign out

Spark 2.4.5 Jobs Stages Storage Environment Executors Zeppelin application UI

Environment

Runtime Information

Name	Value
Java Version	1.8.0_252 (AdoptOpenJDK)
Java Home	/usr/lib/jvm/adoptopenjdk-8-hotspot-amd64/jre
Scala Version	version 2.12.10

Spark Properties

Name	Value
PYSPARK_DRIVER_PYTHON	python
PYSPARK_PYTHON	/opt/conda/default/bin/python
master	yarn-client
spark.app.id	application_1591009784507_0002
spark.app.name	Zeppelin
spark.driver.appUIAddress	/gateway/default/node?host=yarn-cluster-m.c.prashant-gcp-project.internal&port=34587
spark.driver.extraClassPath	:/usr/lib/zeppelin/interpreter/spark/*::/usr/lib/zeppelin/interpreter/zeppelin-interpreter-api-0.9.0-SNAPSHOT.jar:/usr/lib/zeppelin/interpreter/spark/spark-interpreter-0.9.0-SNAPSHOT.jar:/etc/hadoop/conf
spark.driver.extraJavaOptions	-Dfile.encoding=UTF-8 -Dlog4j.configuration=file:///etc/zeppelin/conf/log4j.properties -Dzeppelin.log.file=/var/log/zeppelin/zeppelin-interpreter-spark-shared_process-zeppelin-yarn-cluster-m.log
spark.driver.host	yarn-cluster-m.c.prashant-gcp-project.internal
spark.driver.maxResultSize	960m
spark.driver.memory	1920m

Cluster details – Dataproc – Pras | https://ebnrzrzk3jfkbb6dujpuicu | https://ebnrzrzk3jfkbb6dujpuicu

ebnrzrzk3jfkbb6dujpuicuvgi-dot-us-central1.dataproc.googleusercontent.com/gateway/default/yarn/proxy/application_1591009784507_0002/environment/

prashant-gcp-project > yarn-cluster

Sign out

Spark Properties

Name	Value
PYSPARK_DRIVER_PYTHON	python
PYSPARK_PYTHON	/opt/conda/default/bin/python
master	yarn-client
spark.app.id	application_1591009784507_0002
spark.app.name	Zeppelin
spark.driver.appUIAddress	/gateway/default/node?host=yarn-cluster-m.c.prashant-gcp-project.internal&port=34587
spark.driver.extraClassPath	:/usr/lib/zeppelin/interpreter/spark/*::/usr/lib/zeppelin/interpreter/zeppelin-interpreter-api-0.9.0-SNAPSHOT.jar:/usr/lib/zeppelin/interpreter/spark/spark-interpreter-0.9.0-SNAPSHOT.jar:/etc/hadoop/conf
spark.driver.extraJavaOptions	-Dfile.encoding=UTF-8 -Dlog4j.configuration=file:///etc/zeppelin/conf/log4j.properties -Dzeppelin.log.file=/var/log/zeppelin/zeppelin-interpreter-spark-shared_process-zeppelin-yarn-cluster-m.log
spark.driver.host	yarn-cluster-m.c.prashant-gcp-project.internal
spark.driver.maxResultSize	960m
spark.driver.memory	1920m
spark.driver.port	41819
spark.dynamicAllocation.enabled	true
spark.dynamicAllocation.maxExecutors	10000
spark.dynamicAllocation.minExecutors	1
spark.eventLog.dir	gs://dataproc-temp-us-central1-315784728289-5r6o9pun/205b1cc7-2ada-4aa0-87c3-a25f440a9532/spark-job-history
spark.eventLog.enabled	true
spark.executor.cores	1
spark.executor.id	driver
spark.executor.instances	2
spark.executor.memory	2688m

Cluster details – Dataproc – Pras | https://ebnrzrzk3jfkbb6dujpuicu | https://ebnrzrzk3jfkbb6dujpuicu

ebnrzrzk3jfkbb6dujpuicuvgi-dot-us-central1.dataproc.googleusercontent.com/gateway/default/yarn/proxy/application_1591009784507_0002/environment/

prashant-gcp-project > yarn-cluster

Sign out

spark.executor.id	driver
spark.executor.instances	2
spark.executor.memory	2688m
spark.executorEnv.OPENBLAS_NUM_THREADS	1
spark.executorEnv.PYTHONPATH	/usr/lib/spark/python/lib/py4j-0.10.7-src.zip:/usr/lib/spark/python/:<CPS>{{PWD}}/pyspark.zip<CPS>{{PWD}}/py4j-0.10.7-src.zip
spark.hadoop.hive.execution.engine	mr
spark.history.fs.logDirectory	gs://dataproc-temp-us-central1-315784728289-5r6o9pun/205b1cc7-2ada-4aa0-87c3-a25f440a9532/spark-job-history
spark.jars	file:/usr/lib/zeppelin/interpreter/spark/spark-interpreter-0.9.0-SNAPSHOT.jar
spark.master	yarn
spark.org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter.param.PROXY_HOSTS	yarn-cluster-m
spark.org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter.param.PROXY_URI_BASES	/gateway/default/yarn/proxy/application_1591009784507_0002/
spark.repl.class.outputDir	/tmp/spark8660106897015539719
spark.repl.class.uri	spark://yarn-cluster-m.c.prashant-gcp-project.internal:41819/classes
spark.rpc.message.maxSize	512
spark.scheduler.minRegisteredResourcesRatio	0.0
spark.scheduler.mode	FAIR
spark.shuffle.service.enabled	true
spark.sql.catalogImplementation	hive
spark.sql.cbo.enabled	true
spark.sql.parquet.cacheMetadata	false
spark.submit.deployMode	client
spark.ui.filters	org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
spark.ui.port	0
spark.useHiveContext	true

Cluster details

SUBMIT JOB REFRESH DELETE VIEW LOGS

yarn-cluster

The cluster will be deleted when idle exceeds 1 hour

prashant@yarn-cluster-m: ~ - Google Chrome

ssh.cloud.google.com/projects/prashant-gcp-project/zones/us-central1-f/instances/yarn-cluster-m?authuser=1&hl=en_GB&projectNumber=315784728289

```
prashant@yarn-cluster-m:~$
prashant@yarn-cluster-m:~$
prashant@yarn-cluster-m:~$
prashant@yarn-cluster-m:~$ spark-submit --help
Usage: spark-submit [options] <app jar | python file | R file> [app arguments]
Usage: spark-submit --kill [submission ID] --master [spark://...]
Usage: spark-submit --status [submission ID] --master [spark://...]
Usage: spark-submit run-example [options] example-class [example args]

Options:
  --master MASTER_URL         spark://host:port, mesos://host:port, yarn,
                              k8s://https://host:port, or local (Default: local[*]).
  --deploy-mode DEPLOY_MODE   Whether to launch the driver program locally ("client") or
                              on one of the worker machines inside the cluster ("cluster")
                              (Default: client).
  --class CLASS_NAME          Your application's main class (for Java / Scala apps).
  --name NAME                 A name of your application.
  --jars JARS                 Comma-separated list of jars to include on the driver
                              and executor classpaths.
  --packages                  Comma-separated list of maven coordinates of jars to include
                              on the driver and executor classpaths. Will search the local
                              maven repo, then maven central and any additional remote
                              repositories given by --repositories. The format for the
                              coordinates should be groupId:artifactId:version.
  --exclude-packages          Comma-separated list of groupId:artifactId, to exclude while
                              resolving the dependencies provided in --packages to avoid
                              dependency conflicts.
```

yarn-cluster

The cluster will be deleted when idle exceeds 1 hour

prashant@yarn-cluster-m: ~ - Google Chrome

ssh.cloud.google.com/projects/prashant-gcp-project/zones/us-central1-f/instances/yarn-cluster-m?authuser=1&hl=en_GB&projectNumber=315784728289

```
                            in executors can be accessed via SparkFiles.get(fileName).

 --conf PROP=VALUE          Arbitrary Spark configuration property.
 --properties-file FILE     Path to a file from which to load extra properties. If not
                            specified, this will look for conf/spark-defaults.conf.

 --driver-memory MEM        Memory for driver (e.g. 1000M, 2G) (Default: 1024M).
 --driver-java-options      Extra Java options to pass to the driver.
 --driver-library-path      Extra library path entries to pass to the driver.
 --driver-class-path        Extra class path entries to pass to the driver. Note that
                            jars added with --jars are automatically included in the
                            classpath.

 --executor-memory MEM      Memory per executor (e.g. 1000M, 2G) (Default: 1G).

 --proxy-user NAME          User to impersonate when submitting the application.
                            This argument does not work with --principal / --keytab.

 --help, -h                 Show this help message and exit.
 --verbose, -v              Print additional debug output.
 --version,                 Print the version of current Spark.

Cluster deploy mode only:
 --driver-cores NUM         Number of cores used by the driver, only in cluster mode
                            (Default: 1).

Spark standalone or Mesos with cluster deploy mode only:
```

yarn-cluster

The cluster will be deleted when idle exceeds 1 hour

prashant@yarn-cluster-m: ~ - Google Chrome

ssh.cloud.google.com/projects/prashant-gcp-project/zones/us-central1-f/instances/yarn-cluster-m?authuser=1&hl=en_GB&projectNumber=315784728289

```
                            This argument does not work with --principal / --keytab.

 --help, -h                 Show this help message and exit.
 --verbose, -v              Print additional debug output.
 --version,                 Print the version of current Spark.

Cluster deploy mode only:
 --driver-cores NUM         Number of cores used by the driver, only in cluster mode
                            (Default: 1).

Spark standalone or Mesos with cluster deploy mode only:
 --supervise                If given, restarts the driver on failure.
 --kill SUBMISSION_ID       If given, kills the driver specified.
 --status SUBMISSION_ID     If given, requests the status of the driver specified.

Spark standalone and Mesos only:
 --total-executor-cores NUM Total cores for all executors.

Spark standalone and YARN only:
 --executor-cores NUM       Number of cores per executor. (Default: 1 in YARN mode,
                            or all available cores on the worker in standalone mode)

YARN-only:
 --queue QUEUE_NAME         The YARN queue to submit to (Default: "default").
 --num-executors NUM        Number of executors to launch (Default: 2).
                            If dynamic allocation is enabled, the initial number of
                            executors will be at least NUM.
```